

Program : Diploma in Cyber Forensics and Information security	
Course Code : 6282C	Course Title: Fundamentals of Cyber Security
Semester : 6	Credits: 4
Course Category: Open Elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives

- To understand the Basic Concepts in Cyber Security
- To know about Classification of Cyber Security
- To understand different modalities in cyber security

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basic knowledge on computer concepts.		Introduction to IT Systems Lab	I

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Understand the fundamental concepts of cyber security	14	Understanding
CO2	Understand cyber crimes and cyber forensics	16	Understanding Applying
CO3	Understanding the concepts associated with Ethical Hacking	15	Understanding Applying
CO4	Understanding of cyber security vulnerabilities	13	Understanding Applying
	Series Tests	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3	2					
CO3			2				1
CO4	2						1

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand the fundamental concepts of cyber security		
M1.01	Introduction to Cyber Security	3	Understanding
M1.02	Security threat and vulnerability	3	Understanding
M1.03	Cyber security models	2	Understanding
M1.04	Malicious Software	3	Understanding
M1.05	Denial of Service Attacks.	3	Understanding
Contents: Module 1 Introduction to Cyber Security - Types of Attacks(Active/passive), Goals for Security, Security threat and vulnerability, Cyber security models (the CIA triad, the star model). Malicious Software: Viruses, Worms, Information Theft, Key loggers, Phishing, Spyware Payload Stealthing, Backdoors, Rootkits, Distributed Denial of Service Attacks.			
CO2	Understand cyber crimes and cyber forensics		
M2.01	Introduction to Cyber Crime	4	Understanding
M2.02	Types of Cyber Crime	4	Understanding

M2.03	Crypto currency	3	Understanding
M2.04	Cyber Forensics	5	Understanding Applying
	Series Test – I	1	
Contents : Module 2 Introduction to Cyber Crime – Types of Cyber Crime – Classification of Cyber Criminals – Tools used in Cyber Crime – Challenges – Strategies – Cryptocurrency – Bitcoin – Blockchain - Ransomware. Cyber Forensics Definition – Disk Forensics – Network Forensics – Wireless Forensics – Database Forensics – Malware Forensics – Mobile Forensics – Email Forensics			
CO3	Understanding the concepts associated with Ethical Hacking		
M3.01	Ethical Hacking	2	Understanding
M3.01	Essential terminologies	2	Understanding
M3.02	Hacking windows	3	Understanding
M3.03	Network Hacking and Web Hacking	3	Understanding Applying
M3.04	Password hacking	2	Understanding Applying
M3.05	Malware Scanning	3	Understanding Applying
Contents: Module 3 Ethical Hacking: Essential Terminology, Hacking windows – Network hacking – Web hacking –Password hacking, Malware, Scanning, Cracking.			
CO4	Understanding of cyber security vulnerabilities		
M4.01	Cyber Security Vulnerabilities	3	Understanding
M4.02	System administration	3	Understanding
M4.03	Unprotected Broadband communications	3	Understanding Applying

M4.04	Cyber Security Safeguards	4	Understanding Applying
	Series Test – II	1	
Contents : Module 4 Cyber Security Vulnerabilities-Overview, vulnerabilities in software, System administration, Complex Network Architectures, Open Access to Organizational Data, Weak Authentication, Unprotected Broadband communications, Poor Cyber Security Awareness. Cyber Security Safeguards- Overview, Access control, Audit, Authentication, Biometrics			

Text / References

T/R	Book Title/Author
T1	Deje, Dr.Murugan, “Cyber Forensics”, Oxford University Press, India, 2018.
T2	William Stallings and Lawrie Brown, “Computer Security: Principles and Practice”, Prentice Hall.
T3	Deborah G Johnson, “Computer Ethics”, 4th Edition, Pearson Education Publication.
R1	John W. Rittinghouse, William M. Hancock, “Cyber Security Operations Handbook”, ElsevierPub.
R2	Swiderski, Frank and Syndex, “Threat Modeling”, Microsoft Press.

Online Resources

Sl.No	Website Link
1	https://www.coursera.org/specializations/data-mining
2	http://spoken-tutorial.org/